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Trolling is in the Eye of the Beholder: An Attributional Approach to Predicting Perceptions of Trolling and Vigilantism in Online Games

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Abstract

While perpetuated by the few, trolling in online video games is an increasingly common phenomenon. To understand how trolling behaviors gain momentum and spread throughout a community, this study examined when players perceive trolling and why such perceptions compel retaliatory actions. Participants evaluated staged gameplay that offered multiple auxiliary cues to a player's underlying motivations which included the rules of the video game, the target of the behavior, and the norm of the community. Results suggest that players attribute motivations associated with trolling to others' behaviors by weighing the implications of various auxiliary cues accompanying the behaviors. Additionally, perceptions of trolling mediated participants' reported desire to reciprocate harassment. The current study proposes that a more productive approach to understanding the effects and resulting consequences of trolling may be through exploring players' reactions to behaviors and their attributions of others' motivations.

Keywords

trolling, toxicity, harassment, vigilantism, video game, attribution theory

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Trolls are players that harness the unique affordances of online video game environments to frustrate, anger, and harm the health and well-being of other players (Kowert, 2020). Trolling involves a range of tactics including abusive language and threats of physical harm but also types of harassment more distinct to online spaces and video games such as deliberately losing, cheating, and doxing (Anti-Defamation League, 2019; Lacoma, 2020). Research suggests only a minority of players consistently initiate such behaviors (Bryter, 2020; Cook et al., 2018; Maher, 2016), implying a substantial amount of trolling may be the result of reciprocal processes in which players retaliate against affronts (real or perceived) from others. However, if such reciprocal processes reach a certain threshold, then trolling may become a reinforcing spiral that is eventually woven into the social fabric of a community (Hilvert-Bruce & Neil, 2020).

It is crucial to understand how trolling behaviors gain momentum and spread throughout video game communities (Kwak & Blackburn, 2015). However, trolls can be difficult to identify because they often operate within the rules of the games by exploiting loopholes and repurposing conventional behaviors for malevolent intentions (Kwak et al., 2015). Indeed, these methods differ from more overt forms of aggression which is why recent theorizing has distinguished trolling from typical play by focusing on the intentions behind a behavior. That is, trolling is not determined solely by the classification of a behavior but instead by the perception that a behavior was ostensibly designed to elicit a negative emotion (Kowert, 2020). This *perceived* motivation behind behaviors may be the catalyst that spurs further harassment in the name of retribution (e.g., Cook et al., 2018). Indeed, attribution theories demonstrate that people feel a stronger need to retaliate against a threat and feel justified in doing so when the instigating behavior is inferred as purposefully harmful (Dyck & Rule, 1978; Tedeschi et al., 1974). Therefore, in order to mitigate the spread of trolling, we must examine when players perceive trolling behaviors and why such perceptions compel retaliatory actions.

Although past research has enumerated various trolling behaviors that explicitly express intentions to harm targets (e.g., hate speech), a substantial amount of retaliation likely occurs in more equivocal scenarios in which trolling motivations behind a behavior must be inferred (e.g., voice/text chat is unavailable or filtered/disabled by detection software such as in the game Counter-Strike: Global Offensive [Video game], 2012). In such cases where behavioral motivations are not explicitly made clear, the target of the behavior heavily relies on any available information in the environment (i.e., auxiliary cues) to infer such motivations (Uleman et al., 2008; Winter & Uleman, 1984). Although past research (Bazarova & Hancock, 2010) has pushed to examine how auxiliary cues (e.g., source and message features) in online news and social media affect attributional processes such as blame and condemnation (see DeAndrea & Bullock, 2021; DeAndrea & Walther, 2011), more research is needed to identify the auxiliary cues unique to online gameplay environments that players use to infer trolling motivations given such perceptions likely increase rates of retribution. The current study manipulates three potential auxiliary cues of trolling motivations in prerecorded staged gameplay including whether the action is rewarded and whether it conforms or violates intergroup and community norms. Subsequently, participants' attributions of motivations and perceptions of trolling are examined as a predictor of reciprocal harassment behaviors (i.e., troll vigilantism; Cook et al., 2018).

Trolling

Trolling in online games has been identified as a grave issue by both the game industry and the research community. The seriousness of the issue led the game industry to establish the Fair Play Alliance in 2017, which is a global coalition of over 200 game companies committed to developing healthy gaming communities to fight against trolling behaviors (Canossa et al., 2021). Indeed, the game company EA donated one million dollars to eradicate online trolling in 2019 (Gera, 2019). Alongside these efforts of the industry, researchers have investigated diverse aspects of trolling in online games. Past research on trolling gameplay has primarily relied on theoretical approaches used to explain antisocial behaviors across all online contexts (Cook et al., 2019; Hilvert-Bruce & Neil, 2020; Kordyaka et al., 2020; Kowert, 2020; Monge & O'Brien, 2022; Shen et al., 2020) including the online disinhibition effect (Suler, 2004) and the social identity theory of deindividuation (SIDE; Lea & Spears, 1992). The lack of accountability and deindividuation afforded by anonymity in online spaces is a common and powerful explanation of players' disinhibited negative behaviors. Other antecedents of trolling behaviors in gaming contexts have been identified such as prior victimization (Cook et al., 2018, 2019; Kordyaka et al., 2020), player characteristics (i.e., skill levels and their disparities between players; Shen et al., 2020), team-based gameplay (i.e., conflict within and between groups of players organized into opposing teams; Cook et al., 2019; Shen et al., 2020), personality factors (i.e., the dark triad behaviors; Buckels et al., 2014; Mattinen & Macey, 2018), and norms surrounding trolling (Hilvert-Bruce & Neil, 2020; Kitamura et al., 2021).

Nevertheless, research on trolling still struggles to offer precise theoretical predictions of when and why trolling occurs and persists in video games. That is, research is still unclear on why the dynamics of video games seem to breed trolling at a greater rate than other online spaces where disinhibition, personality traits, and norms can suitably explain such behaviors (e.g., Hilvert-Bruce & Neil, 2020). The issue, in part, lies in the nature and diversity of behaviors offered in video games, alongside the creativity players have shown in appropriating them in unintended ways. This has led researchers to define trolling behaviors in broad and vague terms that produce inconsistent lists of behaviors differing in lengths, antecedents, and classifications. To illustrate, some researchers have examined trolling by common types of behaviors (Cary et al., 2020; Cook et al., 2018) while others have focused on indexing their disrupting effects on gameplay (Monge & O'Brien, 2022; Neto et al., 2017) or negative emotions of targets (Kordyaka et al., 2020). Further confusion exists when comparing trolling with other similar research on the broader topic of "toxicity" which oftentimes leads researchers to use the two terms interchangeably (Kowert, 2020). Thus, while most researchers define trolling behavior as a type of (or all types of) negative behavior in online games, their scopes and conditions vary, leaving researchers without a solid footing to progress our understanding of trolling in video games.

Kowert (2020) addresses the disjointed literature by introducing behavioral intent (or motivation) as an important distinction between negative behaviors perpetrated by trolls, whose only goal is to harm other players, and deviant behaviors that produce harmful outcomes in pursuit of other tangential goals such as gaining an unfair advantage by cheating or contrary play (e.g., Smurfing; Monge & Matthews, 2024). Behavioral intent not only provides a clear and easy classification of negative behaviors, but it also emphasizes the subjective nature of trolling. Indeed, trolling behaviors are specific to each community such that one behavior considered to be trolling in one video game may be innocuous in another. Even in practical terms, trolling only materializes in real gameplay when it is perceived by an alleged victim, regardless of a behavior's original intention. As such, a possible key theoretical predictor of trolling might lie in players' perceived motivations behind a behavior, and not the type of behavior as studied in previous research. Research on attribution theories suggests perceived motivations are ascertained from any available information or auxiliary cues observed with the behavior. Therefore, the current study aims to explore auxiliary cues unique to online video gameplay that can determine whether a player perceives a behavior as trolling. A brief review of attribution research is presented followed by an examination of three cues in online video games that may predict attributions of trolling and subsequent desires to reciprocate harassment behaviors.

The Folk-Conceptual Theory of Behavior Explanation (FCTBE)

Attribution theories propose that we are constantly and automatically trying to mind-read others' thoughts, oftentimes to explain and figure out motivations behind their behaviors toward us (Malle, 2005). According to the folk-conceptual theory of behavior explanation (FCTBE; Malle, 1999, 2004, 2011a), people engage in different psychological processes for explaining behaviors that are identified as intentional versus unintentional, in contrast to the problematic person-situation dichotomies found in prior conceptualizations of the fundamental attribution error (Malle, 2011b but also see Bazarova & Hancock, (2010) and DeAndrea & Bullock, 2021 for a discussion within the field of Communication). A behavior is judged as intentional when the actor presumably believes the behavior will bring about a desired outcome. In such cases, people assume the actor underwent a reasoning process in which they weighed various desires and beliefs before choosing and committing to a behavior (i.e., forming an intention to act). Although the FCTBE claims there are three potential paths for explaining intentional behaviors depending on the available information and goals of the observer, the theory claims that most often people explain observed behaviors by reconstructing the actor's reasoning process that led to the behavior (i.e., "reason explanations"; see Malle, 2022). Therefore, people attempt to take the actor's subjective point-of-view to judge the actor's reasoning, deduce their underlying intentionality for a behavior, and subsequently decide on a response.

FCTBE and Media

Past research has demonstrated that various auxiliary cues in online news reports (e.g., facts relevant to intentionality and citing implicit/explicit bias) can influence the reconstruction of an actor's reasoning for discriminatory behaviors and thus, affect perceptions of blame and judgments of condemnation (DeAndrea & Bullock, 2021). However, this research has mainly explored static descriptions of others' behaviors that are unlike video games that supply behaviors in real-time within fluctuating social interactions. That is, unlike curated news reports that clearly define and focus readers' attention on a fixed number of preselected auxiliary cues, the shifting dynamics of online game play makes it difficult to isolate specific auxiliary cues that are consistently important among players. Yet, it can be argued that there are likely two types of auxiliary cues that guide which motivations are deduced from players' in-game behaviors. Research suggests that people believe situational forces (i.e., extenuating circumstances that facilitate aggressive responses) can affect an actor's reasoning and thus affect their inferred motivation for the actor's behavior. For example, aggressive behaviors that presumably come from unwarranted provocations or defending one's values (i.e., situational forces) are considered less immoral, selfish, and aggressive (Reeder et al., 2002). As such, the FCTBE suggests that situational forces influence perceived motivations because they provide a rationale or structure to observed behaviors. However, in the absence of situational forces, people still seek cues for reasons that rationally support the observed behavior. If an explanation is incompatible with the observed behavior, then it will be discarded for a more congruent one. Oftentimes this requires people to speculate alternative explanations (i.e., rationalize), likely derived from past experiences that shape injunctive and descriptive norms, and weigh them until one matches the auxiliary cues offered in the observation (Malle, 2011a). Therefore, the two cues that likely determine perceived motivations for in-game behaviors are the situational forces and rationalizations that narrow down and suggest possible logical explanations for an observation.

Auxiliary Cues of Trolling in Video games

Situational Force: Behavioral Reward. A situational force is any circumstance that justifies aggressive behaviors in social interactions. In the presence of a situational force, aggressive behavior is seen, in the eyes of an observer, as morally acceptable for the specific conditions the actors find themselves in (Reeder et al., 2002). Situational forces likely help players discern the difference between trolling behaviors and harmless fun because, in essence, aggressive behaviors are inherent and endogenous to most gameplay. That is, if the inferred motivation behind an aggressive behavior is attributed to a situational force such as earning points in a video game (e.g., killing another player in a deathmatch), then players are less likely to perceive such behaviors as personally insulting or having the specific motivation of eliciting a negative emotional response (i.e., trolling). Indeed, the more likely interpretation is that players are earning points in the pursuit of enjoyment, mood repair, and intrinsic need fulfillment

(Rieger et al., 2014). Thus, a situational force that likely influences players' perceived motivations of a behavior is whether the video game rewards the behavior with points.

> *H1*: Participants are more likely to perceive a behavior as trolling when the behavior is not awarded points compared to when the behavior is awarded points.

Rationalizations: Intergroup and Community Norms. As suggested by Kowert (2020), perceptions of trolling are likely culturally specific and shaped by norms that organically arise in different game communities. While expressions of trolling are idiosyncratic to specific gaming contexts, perceptions of the behavior as "normal" should consistently take the form of injunctive and descriptive norms (Cialdini & Trost, 1998). For example, a behavior is viewed as normal when the community approves it and typically reacts positively with social rewards (i.e., injunctive norm). Likewise, a behavior may be considered normal if it occurs frequently in a community and therefore effectively describes a defining characteristic of the community (i.e., descriptive norm). Players who are aware of such injunctive and descriptive norms likely leverage such information when reconstructing another player's reasoning for performing a behavior. Indeed, knowledge that a behavior is consistently denounced and rebuked by others (i.e., injunctive norm) should serve as an auxiliary cue to the behavior's underlying intention. Similarly, another effective auxiliary cue should be whether the observed behavior is frequently performed by trolls who are either self-identified or labeled by others (i.e., descriptive norm). As such, the current study explores two auxiliary cues derived from an injunctive and descriptive norm that are likely common across most team-based video games.

Intergroup Norm. The second auxiliary cue is the group membership (i.e., teammate or opponent) of the targeted player. Research on social gameplay suggests the mere presence of teams can activate evolutionarily ingrained trust norms (i.e., the Group Heuristic; Yamagishi et al., 1999) that lead players to expect more exchanges of positive behaviors from teammates compared to opponents (i.e., an intergroup norm; see Velez, 2018). Targeting a teammate, however, is incongruent with such expectations and thus, likely invokes additional rationalizing about the actor's motivations. Indeed, behaviors from video game teammates that violate these expectations are punished (Velez, 2015), suggesting behaviors that do not mutually benefit teammates (or the team as a whole) are potentially inferred as antisocial due to an injunctive norm surrounding the targeting of teammates (Kowert, 2020; Kordyaka et al., 2020; Monge & O'Brien, 2022; Shen et al., 2020). As such, behaviors targeting teammates are more likely to be rationalized by participants as motivated by trolling compared to targeting an opponent.

> *H2*: Participants are more likely to perceive a behavior as trolling when the behavior targets a teammate compared to an opponent.

Community Norm. The third cue that players may depend upon for rationalizing behavioral motivations is the perceived prevalence of trolling within a community (i.e., a descriptive norm). It is common for a community surrounding a video game to be perceived as generally or regularly toxic (Hilvert-Bruce & Neil, 2020) which can occur when enough trolling has been witnessed, perpetrated, and/or experienced that online forums and discussions label the game and its community as toxic (see Perreault & Lynch, 2022 about “Discourse of Gaming”). Toxic communities are self-perpetuating through several processes including learning specific trolling behaviors from the community and the normalization of such behaviors during gameplay (Kordyaka et al., 2020). According to the FCTBE, if situational forces are not present or clear during gameplay, then players may utilize the broader context and consider such descriptive norms as potential cues to rationalize observed behaviors. That is, players may identify a community norm as a background factor (i.e., a causal history reason, see Malle, 2022) that potentially preceded and contributed to another player’s intention for a behavior. Thus, behaviors are more likely to be inferred as motivated by trolling when players in the community expect to see trolling behaviors.

> *H3*: Participants are more likely to perceive a behavior as trolling when observed in a community with descriptive trolling norms compared to a community without trolling descriptive norms.

Trolling Vigilantism: Reciprocal Harassment Behaviors

Perceived motivations are also critical for predicting targets’ reactions to aggressive behaviors. For example, if someone’s negative behavior was justifiable or brought about by situational forces then people likely blame the circumstances that facilitated the aggression and perceive less hostility on the other’s behalf. In such cases, future harm may be addressed by avoiding similar circumstances which can be achieved without taking action against another. However, when there are no situational forces and/or plausible rationalizations that rule out negative behaviors then people are more likely to perceive the other’s motivations as malicious (Reeder et al., 2002). Under these circumstances, the other person is now perceived as a threat, and people feel compelled to take actions that protect them and restore a sense of safety (Dyck & Rule, 1978; Tedeschi et al., 1974). In video games, this often provokes players to reciprocate harassment as a form of retribution or vigilantism (i.e., “the vigilante troll”; Cook et al., 2018; Ross & Weaver, 2012). Indeed, being the victim of or experiencing trolling is the most commonly cited reason for players’ own trolling behaviors and not solely directed at the original instigator but also in future interactions with new players (Cook et al., 2018). Therefore, the current study proposes that situational forces and rationalizations first shape perceived intentions behind a behavior which subsequently prompts desires to harass in return.

> *H4*: Auxiliary cues surrounding a behavior will first increase perceptions of trolling before subsequently leading to increased desires to reciprocate harassment.

Method

Participants

A total of 485 participants were recruited through the platform Prolific. Participants were English-speaking residents of the United States who were 18 years of age or older and indicated at least one of the following as a hobby: computer games, console games, or handheld console games. Participants who failed any of the eight attention checks (see Measures) were removed ($n=167$, see Table A6). Additionally, 18 participants were removed due to open-ended data that suggested the participant guessed the purpose of the study or was confused about the stimuli (see Tables A1 and A2). Finally, nine participants were removed due to the length of time for study completion (i.e., extreme outliers) and one participant was removed due to missing data. Of the remaining valid 290 cases, 80% indicated they were White, 12% that they were Asian, 12% as Spanish, Hispanic, or Latino origin, 19 Black or African American, two American Indian or Alaska Native, two Native Hawaiian or Pacific Islander, and eight indicated the Other category. The average age was 35 and 63% identified as Male, 33.6% identified as Female, eight identified as Non-binary/Third Gender, and two preferred to self-describe (Genderqueer and Agender). Participants reported playing, on average, 14 ($SD=12.56$) hours of video games in a week. Participants were asked how often they specifically play shooter games compared to other genres on a 7-point Likert scale (1 = Never play, 4 = Play shooting games just as often as other genres, 7 = Play(ed) all the Time), and participants indicated that they play shooter games about as often as other genres ($M=4.18$, $SD=1.79$). Participants rated their overall ability at playing shooting video games near the midpoint ($M=3.85$, $SD=1.75$) of a 7-point Likert scale (1 = Rookie, 7 = Expert). Several potential covariates were measured including participants' prior experience with the video game in the stimuli (i.e., *Overwatch*; Yes/No; 42% indicated yes), self-reported frequency of engaging in "toxic behaviors" in gameplay (7-point Likert scale; 1 = Never, 7 = Every time; $M=1.92$, $SD=1.25$), and perceptions of the focal player in the stimuli being targeted with toxic or inappropriate behaviors by other players in the video (7-point Likert scale, 1 = Not at all, 7 = Very Often; $M=1.79$, $SD=1.34$). Analyses examining these covariates did not change the pattern of results and thus, the simpler analyses are reported. The analyses with covariates can be found at (https://bit.ly/trolling_videogame).

Power Analysis

An a *priori* power analysis was conducted using G*Power (3.1; Faul et al., 2009) to determine the required sample size for the current study. The analysis assumed a small to medium effect size of $f=0.175$ based on a conservative estimate from similar past research (Study 1; Monge & Matthews, 2024). Desired statistical power was set as $1-\beta=.80$ and a significance level of $\alpha=.05$. The analysis indicated that 259 participants were needed for the current study and therefore, the 290 valid cases provided sufficient statistical power.

Procedure

The current study employed a 2 (Behavioral Reward: Points vs. No Points) x 2 (Intergroup Norm: Teammate vs. Opponent) x 2 (Community Norm: Trolling vs. Non-Trolling) between-subjects design. The study was approved by the Internal Review Board at the home institution. The current study used the hero-shooter video game *Overwatch* as the contextual setting and for stimuli development. *Overwatch* is a cartoonish, multiplayer, online, team-based first-person shooter video game. Participants first read a fictitious cover story regarding the purpose of the study. Participants were told that administrators from a community of *Overwatch* video game players in the Midwest were soliciting objective feedback of its members' conduct (see Figure A1). Therefore, participants were expected to give a fair assessment of recorded gameplay to help administrators maintain the quality of their community and provide members enjoyable experiences. To aid in participants' evaluations of gameplay, they were presented with the fictitious community's standards of conduct (see Figure A2). The list of community standards was adapted from *League of Legend*'s list of Behaviors that the Community Rejects in their community code of conduct (Riot Games, 2023). The *League of Legends*'s community standards were chosen because they aligned with the behaviors examined in the current study, the game has a similar team-based format as *Overwatch*, and the community is well-known for its toxic norms. In the current study, the community standards provided examples of six behaviors including harassment, "comms abuse," intent to lose (leaving the game/afk), intent to lose (in-game behaviors), offensive or inappropriate names, and cheating.

The community norm was then manipulated depending on participants' randomly assigned condition (see Table 1) by informing them that members typically ignored or followed these standards (see Community Norm Manipulation and Figure A3). Participants were given more specific information about the focal community members and the match in the recorded gameplay. All participants were informed the community member was randomly selected for evaluation and was playing the character Reinhardt with five other members in an unranked match. Likewise, all participants were given brief descriptions of *Overwatch* and the specific game mode being played in the recording (i.e., team deathmatch; see Figure A4). Depending on participants' condition, some received additional information regarding a "special modification" activated during gameplay in which points were awarded for performing "emotes" "near the place where a teammate/opponent has recently died" (see Behavior Reward Manipulation and Figure A4). All participants were presented with a "Confidentiality Notice" stating that voice communication and any other expression of identifying information was deactivated during gameplay to protect the anonymity of its members and thus, emotes were the only means of communication between players. Additionally, participants were informed that all community members reported not knowing each other (see Figure A5). This information was meant to discourage participants from assuming in-game actions were the result of friends engaging in playful behaviors.

In order to ensure correct identification of the focal player and the targets of emotes (i.e., Group Norm Manipulation section), participants then watched a short video

Table 1. Number of Participants by Condition.

Conditions	Removed	Final *n*
Points-teammate-trolling	20	33
No points-teammate-trolling	69[a]	40
Points-opponent-trolling	13	38
No points-opponent-trolling	15	36
Points-teammate-nontrolling	16	36
No points-teammate-nontrolling	29	33
Points-opponent-nontrolling	16	38
No points-opponent-nontrolling	17	36

Note. Conditions are shorthand for Behavioral Reward, Intergroup Norm, and Community Norm.
[a]A large percentage of participants removed from this condition incorrectly identified the target of the focal behavior as the opponent instead of the teammate (approximately 70% or 48 participants). This is likely due to two reasons. First, there were no points in this condition and therefore, no text was provided to reiterate the team membership of the targeted player (see Behavioral Reward Manipulation). Secondly, as argued in the manuscript, there is likely an intergroup norm that leads people to expect more prosocial behaviors between teammates and therefore, participants may have questioned their memory of the Intergroup Norm Manipulation in this condition and assumed the focal player was engaging in the more normative behavior of targeting an opponent compared to a teammate. As such, more participants were run in this condition to ensure approximately equal cell sizes across conditions.

(27 s) introducing the specific characters for the two teams (Blue and Red) and were subsequently asked to place images of each character into the correct teams. Participants who incorrectly placed any character were then shown the correct placement for all characters into their teams. Participants were then given a brief description of the video's contents before watching recorded gameplay according to their condition (see Stimuli section). Afterwards, participants completed measures of causal attribution for the emotes (i.e., internal versus external; the manipulation check), perceptions of trolling by the focal player, likelihood of reacting with harassment behaviors, and likelihood of reporting the focal player. Finally, participants completed attention checks and reported video game habits along with demographics.

Stimuli

The recorded gameplay stimuli were created using the video game Overwatch [Video game] (2016), a team-based shooter. The two teams in the stimuli had a similar make-up which consisted of a tank (i.e., to mainly absorb damage), assault (i.e., to mainly inflict damage), and support class character (i.e., to mainly heal and boost other team members). The two characters for each type (spread across two teams) were purposefully chosen to look as dissimilar as possible to reduce confusion regarding team membership (i.e., Reinhardt and Winston, Torbjorn and Hanzo, Lucio, and Zenyatta). The match took place in the French castle map (i.e., Château Guillard) which has

Figure 1. Focal player emoting over slain teammate for points.

neutral colors (i.e., grey blocks) and open well-lit spaces for highlighting game characters' actions. The goal of the game mode was to simply kill opposing members (i.e., Team Deathmatch) and each video (2:59 min) consisted of four clips showing the focal player (i.e., Reinhardt) fighting an opposing player before emoting over a dead teammate or opponent. All videos began by showing each team (accompanied with team name) running towards the middle of the map at which point the video takes the perspective of the focal player (i.e., Reinhardt) seconds before the teams engage with each other. Once the perspective of the camera switched to Reinhardt, all participants watched the focal player fight against the opposing team with the assistance of their own team before splitting off and, depending on their condition, either fighting and killing an opponent before emoting over their dead body or witnessing a teammate die in a one-on-one fight with an opponent before emoting over their dead body. The focal behavior was the Uproarious emote for Reinhardt in which he laughs while slapping his knee and leaning on his hammer. This emote was chosen due to its versatility and applicability in various contexts, thereby making it subject to various interpretations depending on the circumstances surrounding its use. Immediately before the emote, the camera panned from behind Reinhardt to face him, giving participants a full view of Reinhardt and the fallen player. Depending on the condition, text appeared awarding the focal player additional points for the emote (See Figure 1). The remaining three clips followed the same format except the clips began with a one-on-one fight between Reinhardt and a single opponent. Additionally, participants were randomly assigned to one of three orders of these clips to reduce potential order effects. Finally, the screen names of the players and the total team points displayed at the top of the screen were blurred out in all videos (see Figure 1). The gameplay footage was recorded using OBS Studio and edited using Adobe Premier Pro. Please visit https://bit.ly/trolling_videogameforstimuli.

Manipulations

Behavioral Reward. Prior to the video, half of the participants were informed that the recorded gameplay featured a "special modification" that awarded points to participants who emoted over players' bodies. For those who saw the emote rewarded with points during the video, text appeared once Reinhardt began the emote that read "LAUGH OVER TEAMMATE 10" (see Figure 1). The text was in Zuume Soft Bold Italic font to match the aesthetic of Overwatch and was edited into the video using Final Cut Pro. In conditions where the emote did not earn additional points, no additional information about a modification was presented and no text appeared when the emote was performed.

Intergroup Norm. Participants either witnessed the focal player emote over the bodies of teammates or opponents. For participants in the behavioral reward condition (see above), the additional information about the modification indicated that the emote earned points when performed over teammates or opponents' bodies. For all other conditions, no additional information was presented and participants either saw the focal player emote over an opponent they defeated or over a teammate whom the focal player witnessed die while in pursuit of an opponent.

Community Norm. Immediately after reviewing the fictitious community's standards of conduct, half the participants were told that an internal report indicated that players in this community mostly ignored or followed the standards. Manipulations of this sort have been found to influence perceived norms, attitudes and behavior (Rhodes et al., 2020). The internal report also indicated community members' reactions to the prevalence of trolling using four emotions. These emotions clarified the injunctive norm surrounding trolling behaviors in the community given research suggests that descriptive norms may tacitly imply a behavior is acceptable due to its frequent occurrence (Cialdini et al., 1990). The four emotions in the internal report were chosen due to their pervasiveness in descriptions of trolling behaviors (see Kowert, 2020). In the non-trolling community norm conditions, participants were informed that players within this community reported that it was uncommon to see such behaviors, did not expect to see them, and hoped they never became a defining characteristic of the community (see Figure A3).

Measures

Attention Checks. Given the number of manipulations and complexity of stimuli in the current study, the participants were asked to respond to eight attention checks to ensure the quality of their responses. Two attention checks instructed participants to pick nonsensical options such as selecting "All my friends are aliens." One attention check asked participants to correctly identify the focal player among all the characters shown in the video, and one attention check asked participants to describe the emote. Another attention check asked participants whether the players knew each other before the match on a 7-point Likert scale (1 = Definitely not, 7 = Definitely yes). Participants were removed if they chose the middle option or above. Participants were also asked using dichotomous variables to indicate whether the emote was awarded points, the team membership of the emote target, and whether the internal report indicated that the majority of community members followed or ignored the standard of conduct.[1]

Manipulation Checks

Attribution Processes. Twelve bipolar items on a 7-point scale from the Revised causal dimension scale (CDSII; McAuley et al.,1992) were used to examine whether the manipulations had the intended effect of evoking attributional processes ($M = 5.25$, $SD = 1.21$, $\alpha = .81$). An exploratory factor analysis with Oblimin rotation was conducted on 12 items, identifying three dimensions with Eigenvalues greater than 1. The first dimension was related to attributing the cause to the player (e.g., "manageable by the player - not manageable by the player"; 6 items), and the other two dimensions were associated with attributing the cause to others (e.g., "under the power of other people - not under the power of other people"; 3 items) or the changeability of the situation (e.g., "stable over time - variable over time"; 3 items). Only the first dimension theoretically captured the attributional process targeted by the three manipulations which should have led participants to blame the focal player for engaging in the behavior instead of any external force (see Manipulation Checks for more detail on this logic). The other two dimensions did not encompass perceptions of the focal player (i.e., only other players or temporality) and thus, were not theoretically relevant to the attributions of interest. These observed dimensions were similar to the original dimensions observed by McAuley et al. (1992). Our first dimension related to attributing the cause to the player and is a combination of McAuley et al. (1992) dimensions of causality and personal control, which the authors found to be strongly positively correlated. The other two dimensions of attributing the cause of others and the changeability of the situation were similar to the dimensions of attributions to external control and stability, respectively, observed by the original authors of the CDSII. Please see Table A3, for full results.

Reporting Player. A single item was created on a 7-point Likert scale (i.e., Extremely unlikely – Extremely likely; $M = 2.24$, $SD = 1.78$) to assess the likelihood of reporting the focal players' behaviors (i.e., "How likely would you have reported them to the game administrators or moderators for perceived harassment (e.g., to kick them out of the game, report them, or ban them)").

Perceptions of Trolling. The FCTBE suggests that perceptions of trolling may depend on how an observer reconstructs and judges an actor's reasoning for a behavior. If there are no extenuating circumstances to justify an actor's behavior, then an observer is more likely to infer the actor knew the behavior would elicit negative emotional responses and thus, chose the behavior with that reason in mind. Past research has demonstrated that such unjustifiable behaviors are perceived as more selfish, immoral, and aggressive (Reeder et al., 2002) and therefore, these three items were used to measure perceived trolling on a 7-point Likert scale with the following prompt "How would you describe the player's (REINHARDT) use of emotes (dances or gestures that express a player's emotion) during the game?". Ratings of the behavior as harsh and friendly (reverse coded) were also included to ensure the behavior was perceived as intentionally eliciting a negative emotional response and not as playful taunting. Finally, an additional item from Reeder et al. (2002) measured whether the focal player's use of the emote was motivated by a desire to win (i.e., 1 = Not at all Motivated to Win, 7 = Very Motivated to Win; reverse coded). Although Reeder et al. (2002) used this item to measure unjustifiable motivations for aggressive behaviors, a desire to win in the current game can potentially justify the use of certain behaviors that increase a player's chance of winning (i.e., the emote to frustrate an opponent). However, a Principal Axis Factoring analysis with Oblimin rotation indicated this item did not load onto the resulting single factor and therefore, was excluded due to a low extracted communality (0.07). The full factor analysis can be found in Table A4. Therefore, perceptions of trolling were measured with five items ($M = 4.35$, $SD = 1.39$, $\alpha = .92$).[2]

Trolling Vigilantism: Reciprocal Harassment Behaviors. The video game harassment scale (Fox & Tang, 2017) and its 11 statements were adapted for the current study. The current version of the video game harassment scale asked participants to take the perspective of a community member and answer how likely they would have reacted to the focal player (i.e., Reinhardt) during gameplay with various in-game harassment behaviors (7-point Likert Scale; Extremely unlikely – Extremely likely). The harassment-related behaviors included making comments about their intelligence or ability, insulting them, making homophobic or racist comments, wishing harm, frustrating them in-game, asking them to leave the game, and unfairly reporting them to moderators or administrators ($M = 1.96$, $SD = 1.01$; $\alpha = .87$; see Table A5).

Manipulation Validity and Analysis Plan

Attributional processes are central to the design of the behavioral reward and intergroup norm manipulations which aim to rule out explanations for a player's behavior that would otherwise shift the blame to an external force and not the player themself. While the hypotheses focused on perceptions of trolling and vigilantism, the two manipulation checks ensured the stimuli had the intended attributional effect to increase confidence that trolling perceptions did not arise from some other psychological processes. Specifically, these two manipulations should have led participants to perceive the cause of the behavior as originating from the player, thereby facilitating a more negative interpretation of the observed behavior (i.e., perceptions of trolling). Likewise, if participants perceived the behavior portrayed in these two manipulations as directly incompatible with the community's standards of conduct (i.e., trolling), then they should also feel prompted to use other formal sanctioning methods such as reporting the player to the game administrators. Although the community norm manipulation was not meant to directly affect causal attributions or reporting behaviors, it was nevertheless included in the manipulation check analyses. For a more in-depth explication of the potential effects of each manipulation on the two checks (i.e., attribution processes and reporting), please see Appendix.

Results

Manipulation Checks

Attribution Processes. The effect of the auxiliary cues on participants' causal attributions was examined in a 3-way analysis of variance (ANOVA) with each cue entered as a fixed factor. Results indicated that participants were more likely to attribute the cause of a behavior to the focal player when the behavior was not awarded points ($M = 5.49$, $SE = .10$) compared to when the behavior was awarded points ($M = 5.03$, $SE = .10$; $F[1, 282] = 10.88$; $p = .001$; $\eta^2 = .037$). Additionally, the cause of a behavior that targeted a teammate ($M = 5.48$, $SE = .10$) was more likely attributed to the focal player compared to a behavior that targeted an opponent ($M = 5.04$, $SE = .10$; $F[1, 282] = 10.51$; $p = .001$; $\eta^2 = .036$). The presence or absence of a trolling community norm did not influence participants' causal attributions, $F(1, 282) = 1.80$; $p = .180$; $\eta^2 = .006$. The behavioral reward and intergroup norm manipulations had the intended effect on attribution processes by encouraging participants to infer the intention behind the observed behavior arose from within the focal player and thus, provide an opportunity for perceptions of trolling to emerge.

Reporting Player. The effect of the auxiliary cues on participants' reporting behaviors was examined in a 3-way analysis of variance (ANOVA) with each cue entered as a fixed factor. Results indicated main effects for the intergroup norm and behavioral reward manipulations in the expected directions. Although participants were unlikely to report the focal player across all conditions (i.e., scores were below the scale midpoint), participants were the least likely to report the focal player when they targeted an opponent ($M = 1.89$, $SE = .14$) compared to a teammate ($M = 2.58$, $SE = .14$; $F[1, 282] = 11.75$; $p < .001$; $\eta^2 = .04$). Participants were also the least likely to report the player if they were awarded points for the behavior ($M = 1.92$, $SE = .14$) compared to no points awarded ($M = 2.55$, $SE = .14$; $F[1, 282] = 9.94$; $p = .002$; $\eta^2 = .034$). There was no main effect of the community norm manipulation ($F[1, 282] = 1.08$; $p = .299$; $\eta^2 = .004$). However, an interaction occurred between the intergroup norm and behavioral reward manipulations ($F[1, 282] = 4.74$; $p = .03$; $\eta^2 = .017$). Simple main effects with Bonferroni corrections revealed that participants were more likely to report the focal player for targeting a teammate ($M = 3.12$, $SE = .20$) compared to an opponent ($M = 1.99$, $SE = .20$) only when no points were awarded for the behavior ($F[1, 282] = 15.70$; $p < .001$; $\eta^2 = .053$). Similarly, reporting the focal player for targeting a teammate significantly differed depending on whether points were awarded ($M = 2.04$, $SE = .21$) or not ($M = 3.12$, $SE = .20$; $F[1, 282] = 13.88$; $p < .001$; $\eta^2 = .047$).

The unexpected interaction effect between the intergroup norm and behavioral reward manipulations was driven by the lack of reporting by participants who witnessed the focal player emoting over a dead opponent for no additional points. Perhaps participants felt the emote in this context was not egregious enough to defy the conduct standards, potentially because the emote is a natural part of the game and logically, was designed for the purpose of taunting opponents. That is, although the behavior may have been perceived as intentionally soliciting a negative reaction out of the target, it did not technically break any rules and therefore, was ineligible or did not warrant reporting.

Hypotheses

Perceived Trolling. The first three hypotheses predicted that observing a behavior that was not awarded points (H1; situational force cue), targeted a teammate (H2; rationalization cue), or occurred within a trolling community (H3; rationalization cue) would separately lead to greater perceptions of the behavior as trolling (i.e., main effects). A 3-way analysis of variance (ANOVA) was used with each auxiliary cue entered as a fixed factor. Results indicated significant main effects for each auxiliary cue in the hypothesized directions, supporting H1-3. Specifically, not awarding points for a behavior ($M=4.69$, $SE=.11$) led to greater perceived trolling compared to when points were awarded ($M=4.03$, $SE=.11$; $F[1, 282]=18.58$; $p<.001$; $\eta^2=.062$). Likewise, targeting a teammate ($M=4.60$, $SE=.11$) compared to an opponent ($M=4.12$, $SE=.11$) increased perceived trolling, $F(1, 282)=9.41$; $p=.002$; $\eta^2=.032$. Finally, observing the behavior in a community with trolling norms increased perceived trolling ($M=4.63$, $SE=.11$) compared to when the behavior was observed in a community without such norms ($M=4.08$, $SE=.11$; $F[1, 282]=13.15$; $p<.001$; $\eta^2=.045$). The interaction between all three auxiliary cues was not significant ($F[1, 282]=0.41$; $p=.523$; $\eta^2=.001$) nor were there any interactions between any two auxiliary cues.

Trolling Vigilantism: Reciprocal Harassment Behaviors. The fourth hypothesis, which predicted that auxiliary cues surrounding a behavior will increase vigilantism by first increasing perceptions of said behavior as trolling, was additionally supported. Given the results indicated main effects for the auxiliary cues, the SPSS macro PROCESS was used to test the indirect effects of each auxiliary cue (separately) on vigilantism trolling through perceived trolling. Results indicated significant indirect effects of each auxiliary cue on vigilantism through increased perceptions of trolling. Specifically, observing a behavior performed for no additional points ($\beta=.20$, $SE=.06$; 95% $CI=0.10$ to 0.325), or targeted a teammate ($\beta=.14$, $SE=.05$; 95% $CI=0.05$ to 0.24), or occurred in a trolling community ($\beta=.17$, $SE=.05$; 95% $CI=0.07$ to 0.29) increased perceptions of the behavior as trolling which then increased participants' desire to reciprocate negative behaviors during gameplay from the perspective of a player within the community (see Tables 2 and 3 for all paths). The full results for all analyses can be found at https://bit.ly/trolling_videogame.

Table 2. Direct, Indirect, and Total Effects of Mediation Analyses.

Variables	β	*SE*	*t*	*p*	95% CI
Direct effects					
BehavReward → HarassBehav	−.11	.11	−1.00	.319	[−.34, .11]
InterGNorm → HarassBehav	.08	.11	.70	.484	[−.14, .30]
CommNorm → HarassBehav	−.17	.11	−1.53	.127	[−.39, .05]
Indirect effects					
BehavReward → TrollPercep → HarassBehav	.20	.06			[.10, .32]
InterGNorm → TrollPercep → HarassBehav	.14	.05			[.05, .24]
CommNorm → TrollPercep → HarassBehav	.17	.05			[.07, .29]
Total effects					
BehavReward → HarassBehav	.09	.12	.75	.452	[−.14, .32]
InterGNorm → HarassBehav	.21	.12	1.83	.069	[−.02, .45]
CommNorm → HarassBehav	.00	.12	.03	.973	[−.23, .24]

Note. CI = confidence interval; BehavReward = Behavioral Reward; InterGNorm = Intergroup Norm; CommNorm = Community Norm; TrollPercep = Troll Perception; HarassBehav = Harassment Behavior.

Table 3. Individual Paths (i.e., a and b) of Mediation Analyses.

Models and paths	β	*SE*	*t*	*p*	95% CI
Behavioral reward					
BehavReward→TrollPercep	.49	.16	4.33	.000	[0.37, 1.00]
TrollPercep→HarassBehav	.41	.04	7.24	.000	[0.21, 0.38]
Intergroup norm					
InterGNorm→TrollPercep	.36	.16	3.07	.002	[0.18, 0.81]
TrollPercep→HarassBehav	.38	.04	6.97	.000	[0.20, 0.36]
Community norm					
CommNorm→TrollPercep	.42	.16	3.68	.000	[0.27, 0.90]
TrollPercep→HarassBehav	.41	.04	7.39	.000	[0.22, 0.38]

Note. CI = confidence interval; BehavReward = Behavioral Reward; InterGNorm = Intergroup Norm; CommNorm = Community Norm; TrollPercep = Troll Perception; HarassBehav = Harassment Behavior.

Discussion

Trolling is a common concern in online video game communities, yet little is known about how it emerges and propagates through gameplay. Although research on trolling has cataloged disruptive behaviors (see Kowert, 2020) and examined players that initiate such behaviors (Buckels et al., 2014; Mattinen & Macey, 2018), there are no clear theoretical approaches that address the unique and dynamic nature of social gameplay or how perceptions of trolling may materialize across multiple exchanges. Indeed, trolling does not occur in a vacuum nor are players with inherent trolling motivations the only catalysts for such behaviors to develop. The current study used the FCTBE (Malle, 1999) as a foundation to examine how interpretations of others' behaviors may contribute to the genesis and proliferation of harassment in gameplay. Specifically, the study examined three auxiliary cues players may use (i.e., one situational force and two rationalizations) to infer trolling to in-game behaviors and consequently, whether such attributions incited vigilantism (i.e., reciprocal harassment) in return.

Results suggested that players attribute trolling to others' behaviors by weighing the implications of various auxiliary cues appearing alongside the behavior. Three auxiliary cues were manipulated and examined in the current study including whether the video game condoned the behavior (presence/absence of points as a situational force) and whether the group membership of the targeted player (i.e., injunctive norm) and the community norm surrounding the behavior (i.e., descriptive norm) were consistent with a trolling explanation (i.e., rationalizations of the behavior). The effect of each cue appears to be additive such that behaviors occurring in a trolling community, targeting a teammate, and earning no points are perceived as the harshest trolling. Thus, the results demonstrated that perceptions of trolling aligned with predictions derived from the FCTBE and additionally suggested that players' interpretations of each auxiliary cue are less likely to switch or change depending on other cues. For example, the behavior was perceived as more problematic when occurring in a trolling community compared to a non-trolling community, regardless of whether the video game awarded points and/or an opponent was targeted (both of which are less consistent with a trolling explanation). Likewise, the behavior was considered more problematic when targeting a teammate compared to an opponent, regardless of whether the behavior occurred in a non-trolling community and/or was awarded points. Finally, the behavior was perceived as more problematic when no points were awarded compared to awarded points, regardless of whether it occurred in a non-trolling community and targeted an opponent. These perceptions of trolling then mediated participants' reported desire to harass the instigating player (i.e., troll vigilantism). Specifically, the more trolling participants perceived in the players' behaviors, the more likely they were to indicate reacting with harassment behaviors commonly associated with trolling.

Theoretical Implications

It can be difficult for players to accurately reconstruct the motivations of others in chaotic and fast-paced gameplay. Indeed, in these environments, players with varying relationships employ a wide range of skillsets and strategies (e.g., Wright, 2019). The FCTBE suggests that perceptions of trolling may reside in the midst of this chaos when players efficiently, but perhaps ineffectively, appraise the meaning of others' behaviors through auxiliary cues.

The current study examined three potential cues that players may commonly utilize in their assessment of motivations. Results suggest that earning points for a behavior may be considered a situational force that facilitates the attribution of non-trolling motivations by providing players with an easy moral justification for the behavior. As for the group membership of the targeted player and the community norm surrounding the behavior, the results suggest players use these injunctive and descriptive norms as cues to rationalize the observed behavior (i.e., rationalization cues). Perhaps targeting a teammate infers trolling because it violates a cognitive heuristic and injunctive norm that encourages players to expect and reciprocate prosocial behaviors with teammates (Velez, 2015). That is, players may rationalize targeting a teammate as willfully contradicting a normative behavior which may infer a motivation to elicit a negative emotion from the target. The community norm surrounding the behavior was manipulated by providing participants with an ostensible "internal report" from the community that stated members' perceptions of trolling prevalence. When participants read that members reported seeing or not seeing trolling behaviors (i.e., descriptive norm), they likely assumed the members had previously witnessed the behavior featured in the video (i.e., laughing emote) and had already deemed it as trolling or non-trolling. As suggested by Kowert (2020), the internal report likely provided a culturally specific definition of trolling behaviors and thus, it was used as an auxiliary cue of the focal player's underlying motivations. Overall, the results demonstrate that in the absence of explicitly clear motivations, players search for traces of a behavior's impetus and will resort to a quick analysis of available information.

Each auxiliary cue examined in the current study could point to opposite conclusions depending on the rules of the video game, the target of the behavior, and the norm of the community. As predicted, the capricious nature of inferring motivations subsequently predicted players' own desire to engage in negative behaviors. The current study proposed that people will retaliate against behaviors perceived as trolling, perhaps because trolling is construed as a personal threat which may evoke a need to protect oneself. This suggests trolling may emerge organically during gameplay, even when all players are well-intentioned in their behaviors. That is, all that is required for trolling to emerge among typically friendly players is the misinterpretation of one or more auxiliary cues during gameplay. Indeed, there may be playful or innocuous motivations behind players' behaviors that, at least, superficially resemble trolling. For example, players who are new to a community may not understand its norms and engage in behaviors that are culturally inappropriate or may attempt to initiate a relationship through facetious behaviors (Cook et al., 2018). Without clear and direct communication regarding the motivation behind these behaviors (e.g., verbal trolling; Cook et al., 2018), they run the risk of being perceived as trolls. As a result, this may trigger trolling behaviors particularly given a player's retaliation is likely to be construed as initiating trolling behaviors by the originally friendly actor.

While attributions of trolling may provide insight into when trolling occurs, it may also help explain the lack of aggression found in some competitive contexts. The frustration of competence-impeding gameplay has been shown to increase aggression in solo players (Przybylski et al., 2014), and therefore, competition with others should theoretically also lead to such negative effects, if not more, given the potential for inferring antisocial motivations to an opponent's aggressive in-game behaviors. However, research has shown that competitive gameplay does not exceed the aggressive effects of playing alone in contrast to cooperative play which significantly decreases negative effects (Study 2, Velez et al., 2016) and increases prosocial ones (Ewoldsen et al., 2012; Greitemeyer & Cox, 2013; Greitemeyer et al., 2012; Kim et al., 2022; Shoshani & Krauskopf, 2021; Velez, 2015; Velez et al., 2014; Verheij et al., 2020). In light of the current study, this may occur because participants are not attributing malicious or ill-intent to their opponents' behaviors which does not contribute to aggressive behaviors beyond any evoked by competence-impeding frustration. That is, perhaps when reconstructing their opponent's reasoning for in-game behaviors, the logical conclusion is that their competitor is not motivated by a desire to elicit negative emotional reactions but instead to participate in the study for the designated incentive (e.g., extra credit or money). Future research can examine this implication by manipulating perceived motivations of opponents during in-lab experiments (e.g., benign or toxic) and examine if inferred motivations contribute to players' subsequent aggressive behaviors.

The current study also suggests future research on attributional processes should consider the various situational forces commonly found in online environments that provide users with easy and convenient "reason explanations" for others' behaviors, particularly antisocial ones. Unlike real-world interactions where the subjective rewards and benefits of behaviors are not easily accessible, online environments that employ gaming elements such as point systems offer clear and objective evidence of a behavior's utility. Indeed, most video games offer other situational forces such as narratives that represent players' antisocial behaviors as a moral necessity to save and protect others (Hartmann & Vorderer, 2010). Future research should explore situational forces common to virtual environments and gaming spaces that provide clear structures to users' and others' behaviors.

The current study's attributional approach of identifying auxiliary cues that shape players' interpretations and reactions to online behaviors may provide an avenue for exploring antisocial behaviors in other online spaces. Virtual worlds such as the Metaverse are becoming more popular due, in part, to their mixture of verbal communication and non-verbal behaviors, similar to video games. Users in these virtual worlds will likely rely on the attribution processes examined in the current study to interpret and react to perceived antisocial behaviors from others. However, there are spaces in these virtual worlds which lack the structure found in video games that provide point systems, limited behaviors pertinent to gameplay, and the social architecture of teams, guilds, and clans. While the current study utilized aspects of the FCTBE to exemplify the utility of an attributional approach to trolling behaviors in video games, future research into these unstructured and unfettered virtual worlds should utilize the full breadth of FCTBE that includes the four modes of explanation for unintentional and intentional behaviors (Malle, 1999) and the model of the folk concept of intentionality (Malle & Knobe, 1997).

Live streaming is another online space that can benefit from the current findings. The passive observation of watching recorded gameplay in the current study (although see Limitations section) is similar to the increasingly popular activity of watching streamers play video games. As such, viewers are likely processing auxiliary cues present in a streamer's gameplay and engaging in similar attributional processes. Such processes may influence viewers' injunctive and descriptive norms of a game's community and, in some cases, may lead to the formation of such norms for viewers without any personal experience with a game or community. Furthermore, specific behaviors that elicit emotional reactions from live streamers may also imbue those behaviors with trolling or toxic meaning. Future research should examine if and how live streaming contributes to the injunctive and descriptive norms of gaming communities and the behaviors unique to them.

Practical Implications

Trolling has been shown to be a reciprocal and thus, a self-perpetuating phenomenon (Cook et al.,2018; Ross & Weaver, 2012). That is, trolling seems to be contagious (Shen et al.,2020). Many interventions are designed to alleviate trolling by punishing offenders, with the hope that consistent and harsh repercussions will deter players' urge to troll others. However, the current study suggests alternative means for reducing trolling by disrupting the chain of trolling behaviors (i.e., focusing more on the targets of trolls). Specifically, identifying such auxiliary cues may allow developers to nudge players towards inferring non-trolling motivations behind behaviors or at least, make players question a trolling interpretation of another's gameplay. For example, providing a trivial award for behaviors (or even behaviors that are unequivocally trolling) may serve as a situational force that makes the behavior morally acceptable, regardless of the actor's actual motivation. Although it is counter intuitive to reward a potentially aspiring troll, the current study suggests that if targets do not perceive the behavior as trolling, then such behaviors may be less likely to spread. However, in cases where targets incorrectly infer trolling motivations, an opportunity for an actor to clearly state their intentions through either direct communication or prearranged text options may placate or rectify such scenarios (see Monge & Matthews, 2024 for examples).

On a larger scale, companies may also want to invest in public relations campaigns that address perceived trolling norms surrounding their communities. This is in contrast to how companies have historically marketed their games as communities that invite these types of behaviors (Morrissette, 2020). Indeed, the current study suggests that behaviors are more likely to be inferred as trolling if they occur within a community known for trolling. If companies employ effective strategies for reducing trolling behaviors but retain a toxic image, then it may be difficult to maintain low levels of trolling. Companies may also mitigate the spread of trolling by accurately identifying auxiliary cues that lead to perceptions of trolling in their communities. For example, companies can train human referees, community administrators, and new players on which specific behaviors are likely construed as trolling (Blackburn & Kwak, 2014) such as targeting a teammate with the laughing emote in the current study. Indeed, even if players believe they are being light-hearted when using such behaviors, the current study suggests intergroup norms surrounding expected teammate behaviors are precarious and should be protected.

The results surrounding which behaviors participants would report to administrators may provide an additional suggestion for reducing the prevalence of trolling within a video game community. There was a discrepancy between which behaviors prompted participants to report the focal player versus engage in vigilantism. Although participants were unlikely to report emoting over a dead opponent's body when no points were awarded, they were comfortable retaliating with their own harassment for the same behavior. As suggested above, participants may have thought the behavior did not satisfy the criteria for reporting or that administrators would have perceived the behavior as just "part of the game." As such, this may encourage vigilantism, particularly when players feel a certain behavior was performed with the intention of eliciting negative emotions but is technically not reportable. That is, players may feel the only means of remediation is to reciprocate with their own harassment. However, reporting systems can deviate from listing types of problematic behaviors and focus more on players' perceived intentions behind behaviors they wish to report. This may involve including more culturally specific examples of reportable behaviors that consider the perceived intent behind behaviors as agreed upon by the community.

The attribution approach advocated in the current study can be useful in other online spaces that mainly revolve around verbal communication such as social media. Research on cyberbullying suggests that online spaces present greater barriers and obstacles for interventions by "cyber-bystanders," largely due to the equivocality of social media where information regarding the relationship and context of those involved is sparse (Dillon & Bushman, 2015). As such, online users struggle to accurately identify cyberbullying because they fail to identify such instances as an emergency (Dillon, 2016), a crucial step in the Bystander Intervention Model (Latané & Darley, 1968). However, research that identifies auxiliary cues that help online users more accurately reconstruct another's reasoning for a post may help determine when and why cyber-bystanders decide to intervene on behalf of victims. For instance, the current study adds to a large literature demonstrating the importance of injunctive and descriptive norms for influencing people's interpretations and reactions to others in various health, environmental, and socio-cultural issues (Rhodes et al., 2020). A greater understanding of these norms as auxiliary cues of cyberbullying intentions may help boost interventions not only in social media but also in gaming contexts (see Cook, 2021).

Limitations

The current study's stimuli utilized prerecorded staged gameplay that featured actors following a script. Although this allowed for greater experimental control over the manipulations and reduced confounding variables in a typically chaotic environment (i.e., higher internal validity), it also affected the results' external validity. For example, participants did not engage in actual gameplay and thus, were passive observers of the social interactions and contexts surrounding the focal behavior. Although participants reported trolling vigilantism from the perspective of a community member within the recorded gameplay, it is still unclear whether players will similarly perceive and interpret the proposed auxiliary cues while actively engaging in frenetic social gameplay. Indeed, the proposed processes can be automatic and implicit (see Malle, 2005), suggesting the cognitive load and emotional states of playing versus observing may influence how players reconstruct another's reasoning for in-game behaviors. As such, the current findings may overestimate the presence and role of the proposed attributional processes. Future research should examine the roles of relevant processes that are solely found while actively playing. These include flow states that can narrow a player's focus (Weber et al., 2009), satisfaction of intrinsic needs that profoundly connect gameplay to self-relevant processes such as self-esteem (Ryan et al., 2006), and the demands of gameplay on players' attentional resources (i.e., cognitive, emotional, social, and physical; Bowman, 2021).

Additionally, the stimuli purposefully excluded verbal communication between players to ensure the behaviors were ambiguous. As mentioned above, communication is often the vehicle for trolling or is used to express trolling intentions behind in-game behaviors. Although research has demonstrated that attribution processes still account for perceptions of blame and toxicity when players' motivations are made explicitly clear with verbal text (Monge & Matthews, 2024), the meaning and interpretation of the auxiliary cues currently examined may be different or play a lesser role when behavioral motivations are explicitly communicated.

Conclusion

Some trolls are resolute in their intentions to behave negatively. Such trolls may have a predisposition (Buckels et al., 2014) or harbor antisocial motivations (e.g., attack trolls, Cook et al., 2018) that likely persist in spite of interventions. However, most trolling presumably comes from the aggregate of "occasional offenses" carried out by "normal players" (Shen et al., 2020, p. 2). Indeed, this is consistent with research suggesting that most players do not initiate trolling but instead are triggered after being victimized. Yet, it is unclear how so many players encounter trolling when so few are instigating such behaviors. This study offers one potential explanation by suggesting that players can misattribute trolling to in-game behaviors which then provokes further harassment (i.e., vigilante trolling; Cook et al., 2018).

The current study suggests the mere suspicion of trolling may play a significant role in the initiation and escalation of trolling behaviors in online spaces. Indeed, the harassment and aggression associated with trolling are not new and occur in offline spaces as well (Cook, Tang, & Lin, 2023; Ortiz, 2020; Perreault & Lynch, 2022). However, trolling in online video games likely hinges on attributional processes to a far greater extent than real-world scenarios due to the poor communication (e.g., text and/or limited voice chat) but behaviorally rich environment found in games. Future research should pay greater attention to attributional processes that predict perceptions of and reactions to trolling.

Appendix

Welcome and thank you for participating today.

WHO WE ARE

We are a community of video game players that offer members from across the Midwest a wide range of games, organized events, and access to other players that vary in expertise, skill levels, and interests. As administrators of this group, our goal is to encourage a high level of player engagement in competitive and casual gameplay. In order to ensure enjoyable experiences and maintain community quality, we solicit objective assessments of our members and their gameplay.

PURPOSE OF THIS SURVEY

We want to ensure that members receive a fair assessment and thus, in addition to internal reviews, we use platforms such as Prolific to recruit people outside the community to give objective evaluations of our players. Today, you will watch recorded gameplay of one community member playing an unranked match of OVERWATCH with other members of the same community. We hope to get your honest reactions to this player's behaviors.

Figure A1. Cover story.

OUR COMMUNITY

In order to make an informed decision, we want to give you more information about our community standards and the results of an internal review regarding their use within the community.

Our Community Standards reject the following behaviors:

- Harassment
  - *In-game trolling, emotional abuse, bullying. Out-of-game physical harm, doxxing*
- Comms Abuse
  - *Offensive language, hateful speech, sexual harassment*
- Intent to Lose - Leaving the Game/AFK
  - *Disconnecting, AFKing, refusing to play*
- Intent to Lose - In Game Behaviors
  - *Helping opponents, sabotaging team*
- Offensive or Inappropriate Names
  - *Player names, account names, group names*
- Cheating
  - *Hacks, bug abuse, account sharing, account boosting*

Figure A2. Fictitious standards of conduct.

INTERNAL REVIEW RESULTS:

An internal review shows that players mostly ignore these standards. Current members report that it is common to experience behaviors that make them feel angry, frustrated, annoyed, and stressed-out. Many members say these negative behaviors are now expected and have become a defining characteristic of the community.

INTERNAL REVIEW RESULTS:

An internal review shows that players mostly follow these standards. Current members report that it is uncommon to experience behaviors that make them feel angry, frustrated, annoyed, and stressed-out. Many members say they don't expect to see negative behaviors and hope such behaviors never become a defining characteristic of the community.

Figure A3. Community norm manipulation.
Note. Script in trolling condition (Top) and non-trolling condition (Bottom).

PLAYER EVALUATION

A single player from our community was randomly selected for evaluation. The video shows the community member playing the character REINHARDT from the video game OVERWATCH with FIVE other members in an UNRANKED match.

GAME INFORMATION: Overwatch is a team-based shooter that includes a cast of unique heroes, each with their own powers and abilities.

GAME MODE: In this recorded gameplay, six players across two teams (3v3) are attempting to eliminate each other to earn points in a game mode called Team Deathmatch.

SPECIAL MODIFICATIONS: Modifications or "mods" that alter the settings and rules of a match are a common resource to keep our community members engaged and entertained. This particular match featured a **unique modification that awards additional points to teams who perform "emotes" (dances or gestures that express a player's emotion) near the place where a teammate has recently died.*

Figure A4. Behavior reward manipulation for emoting over teammates.

CONFIDENTIALITY NOTICE (Please read): To protect the privacy of our community members, we do not allow the expression of identifying information during matches that are randomly selected for evaluation, and thus, you will not observe voice/text chat or personalized usernames. The only way players are allowed to communicate is through "emotes". Additionally, players are required to indicate whether they personally know the other players. For today's video, **none of the players reported knowing each other.*

Figure A5. Confidentiality notice.

Table A1. Participant Knows That the Gaming Community is Not Real.

No.	Comments
1	It appears to be a controlled environment.
2	I play the game regularly and the gameplay depicted was extremely unrealistic to the point where it seemed to be completely scripted.
3	The players looked like bots
4	The game play looks fake and simulated

Table A2. Confusion Regarding Stimuli.

No.	Comments
1	Nothing wrong with the study, just now I am questioning if my attention checks were good on whether they knew each other. . . I guess you're getting free work if this goes through
2	It was too fast to collet any thoughts or recognize characters to accurately answer these questions
3	I was confused about what was going on during the game. I just saw that the player laughed at the teammate for dying at least twice in the game. I couldn't understand which players were on what team. I wish they were in different colors to be more clear. Watching them on the screen was a lot harder to identify the different players/characters than expected.
4	I am not familiar with this game so it was very difficult for me to make sense of what I was watching as well as the mechanisms of the game
5	Overall, it seemed like a pretty normal online game experience, swearing included. The swearing though was typical in-game swearing and not generally directed at an individual as far as I could tell.
6	I'm not familiar with *Overwatch*. So, it was difficult to follow at times
7	I didn't understand much of what was going on, since it was happening very fast.
8	It was hard to tell what was happening because I've never played *Overwatch* before, so maybe a description of what I'm looking for?

Table A3. Factor Loadings for Attribution Processes.

Component	1	2	3
Manageable by the player - Not manageable by the player	.797		
The player can regulate - The player cannot regulate	.770		
Over which the player has power - The player has no power	.725	−.323	
Inside of the player – Outside of the player	.647		
Something about the player – Something about others	.575	−.447	
Reflects an aspect of the player - Reflects an aspect of the situation	.460	−.334	.399
Other people can regulate - Other people cannot regulate	−.367	.779	
Over which others have control - Over which others have no control		.715	
Under the power of other people - Not under the power of other people		.676	
Permanent - Temporary			.658
Unchangeable - Changeable	−.459		.474
Stable over time - Variable over time			.393

Table A4. Factor Loadings for Perceptions of Trolling.

Component	Loadings
Friendly - Hostile	.886
Moral - Immoral	.844
Not at all aggressive - Very aggressive	.826
Not at all harsh - Very harsh	.821
Unselfish -Selfish	.781
Not all motivated to win - Very motivated to win	-

Table A5. Factor Loadings for Reciprocal Harassment Behaviors.

Component	Loadings
Would have wished for or threatened harm to someone close to them	.828
Would have wished for or threatened harm to them	.768
Would have made comments about their intelligence	.702
Would have said curse or swear words directed at them	.597
Would have made homophobic comments or used homophobic slurs	.696
Would have been rude to them or insulted them	.676
Would have made comments about their abilities to play	.537
Would have asked them to leave the game	.515
Would have made racist comments or used racist slurs	.487
Would have done something in-game to frustrate them	.430
Would have threatened to report them or reported them to the game administrators or moderators unfairly	.396

Table A6. Number of Participants Who Failed Attention Checks by Conditions.

Conditions	Both of the nonsensical attention checks	Identifying the focal character	Whether the players knew each other	Whether the emote was described correctly	Whether points were awarded	Team membership of the emote target	Identifying the community norm
No points-teammate-toxic community	4	1	9	5	11	48	14
Points-teammate-toxic community	2	1	1	3	8	11	3
No points-opponent-toxic community	3	0	0	2	4	1	6
Points-opponent-toxic community	5	1	0	1	3	1	8
No points-teammate-nontoxic community	2	0	1	3	5	17	5
Points-teammate-nontoxic community	2	0	3	0	3	5	2
No points-opponent-nontoxic community	2	1	2	2	5	2	5
Points-opponent-nontoxic community	4	0	1	2	4	0	4
Total[a]	24	4	17	18	43	85	47

Notes. Conditions are shorthand for Behavioral Reward, Intergroup Norm, and Community Norm.
[a]Participants were removed for failing any of the above and thus, the number removed ($n = 167$) is smaller than the totals shown here due to some participants failing multiple attention checks.

Rationale for Manipulation Checks

Attribution Processes. The behavioral reward and intergroup norm manipulations were designed for participants to infer and attribute the focal player's behaviors to trolling by, in part, removing any obvious situational or external explanations of the behavior that could shift the blame or cause to an external force and not the player themself. For example, the cause of a behavior that earns additional points or distracts an opponent should clearly stem from an external reason such as the game's rules (i.e., the point system) or the game's goals (i.e., defeating the opposing team). However, performing a behavior with no clear benefit or strategic advantage in team-based competition (i.e., a behavior that does not earn additional points or targets a teammate) should suggest the focal player had a personal or internal reason for choosing the behavior. As such, the behavioral reward and intergroup norm manipulations should have led participants to perceive the cause of the behavior as originating from the player, thereby facilitating a more negative interpretation of the observed behavior (i.e., perceptions of trolling). The community norm manipulation was not meant to directly affect causal attributions but serve as contextual information that may inform participants' reconstruction of the focal player's reasoning for the behavior. Nonetheless, it is important to determine whether causal attributions were present for the community norm manipulation and thus, it was included as part of the manipulation check.

Reporting Player. Perceptions of trolling heavily rely on witnessing behaviors that are interpreted as problematic and outside of typical gameplay. When players are motivated to redress such situations, reciprocating harassment is only one option available to them. Most video games offer players formalized and sanctioned methods of addressing trolls to discourage retributive justice. These commonly appear as reporting systems in which players can identify a problematic player who violate standards of conduct and send a brief report to game administrators with a short description of the observed behavior and/or by clicking preset options (Shen et al., 2020). Many such reporting systems directly mention trolling as grounds for submitting a report and although the definition of the term "trolling" likely differs across players, if participants perceive the observed behavior in the current study as directly incompatible with the community's standards of conduct, then they should also feel prompted to use other formal methods such as reporting.

The intergroup norm manipulation should directly conflict with at least one of the ostensible community standards, specifically the "Harassment" standard that includes "in-game trolling, emotional abuse, [and] bullying." For example, observing the focal player performing the behavior (i.e., the emote) over the body of teammates should lead to more reporting than emoting over opponents. However, if the game explicitly allows and even encourages emoting over teammates or opponents by awarding points (i.e., the behavioral reward manipulation) then the behavior likely becomes compliant with the conduct standards. Therefore, rewarding the emote with points should lead to less reporting than the same behavior without awarded points. It is unclear if and how the community norm manipulation may impact participants' likelihood of reporting. For example, reporting players in a community where such behaviors are commonplace may be perceived as ineffective and useless. However, participants in the current study were ostensibly reviewing gameplay for a group of administrators which may have increased participants' proclivity to report a behavior. That is, participants may have believed this group of administrators take reporting very seriously, regardless of the community norm, and therefore, feel the reporting system is worth using.

Data availability statement

The data are available from the corresponding author upon reasonable request.

Declaration of Conflicting Interests

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Notes

1. Due to a survey flow error in Qualtrics, approximately half of the participants ($n = 147$) received only one of the three attention check questions regarding the manipulations (i.e., the community norm manipulation). A dichotomous variable representing whether the participants received all three attention checks was entered as a covariate in all analyses. The covariate was not significant nor did the results change as a result and thus, the simpler analyses without the covariate are presented.
2. The definition of the term "trolling" likely differs across players and therefore, this term was not used in the measure of perceived trolling.

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